

Copula Entropy

Theory and Applications

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- Lecture 7: Generalizations
- **Lecture 8: Real applications and softwares**

Lecture 8: Real applications and softwares

1 Real applications

- Astronomy
- Physics
- Chemistry
- Geology
- Biology
- Medicine
- Social sciences
- Engineering

2 Softwares

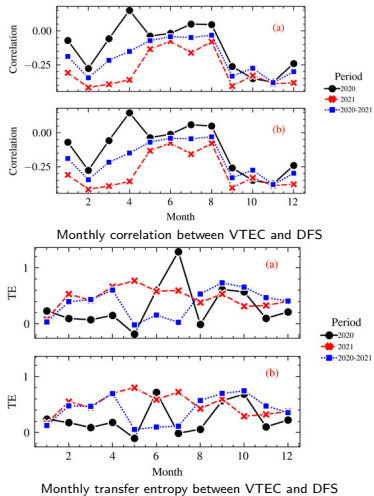
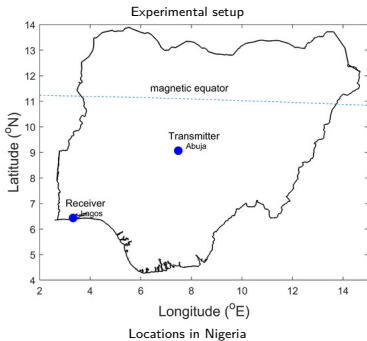
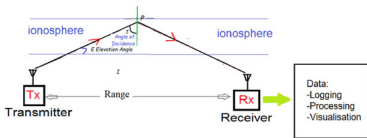
- Official
- Third-party

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- 赤道电离层特性随季节变化分析
 - 基于 TE 分析赤道地区高频微波信号传输中 TEC 和 DFS 之间的非线性关系³

³ Aderonke Akerele et al. "Complexity and Nonlinear Dependence of Ionospheric Electron Content and Doppler Frequency Shifts in Propagating HF Radio Signals within Equatorial Regions". In: *Atmosphere* 15.6 (2024), p. 654. DOI: 10.3390/atmos15060654. URL: <https://www.mdpi.com/2073-4433/15/6/654>.

Table 1 Demographic characteristics of study population



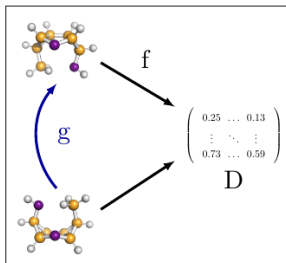
热学

- 微热管设计和制造
 - 提出了一种基于 CE 和神经网络模型相结合的微热管设计和制造参数预测方法⁶
- 循环流化床锅炉工艺参数优化
 - 结合 CE、加权融合树模型和多目标树结构估计器的循环流化床锅炉工艺参数优化方法⁷

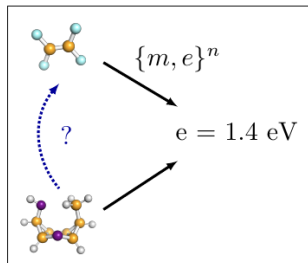
⁶李勇 et al. “基于一维卷积残差网络的微热管几何结构及制造工艺参数预测”. In: 中国科学: 技术科学 55.2 (2025), pp. 281–294, 张靖昊. “基于机器学习的微热管结构及工艺参数设计系统开发”. 硕士学位论文. 华南理工大学, 2024.

⁷Yunpeng Ma et al. “Improve thermal efficiency and reduce NOx emission of circulating fluidized bed boiler based on multi-objective optimization framework”. In: *Thermal Science and Engineering Progress* (2025), p. 103753. ISSN: 2451-9049. DOI: <https://doi.org/10.1016/j.tsep.2025.103753>. URL: <https://www.sciencedirect.com/science/article/pii/S2451904925005438>.

CASE 3: Symmetry of molecules by Harvard and Basel



(a) Rotational symmetry transformation.



(b) Unknown symmetry transformation.

Figure 1: **Left:** a molecule is rotated by g admitting an analytical form. The distance matrix D between atoms is calculated by a known function f and remains unchanged for all rotations. **Right:** n samples $\{m, e\}^n$ where m is the molecule and e the bandgap energy. These samples approximate the function f whereas the class of functions g leading to the same bandgap energy is unknown.

- 土地干旱度分类
 - 考虑蒸散过程的土地干旱度分类¹⁸
- 北欧山地陆气耦合分析
 - 利用 CE 分析了芬瑟两个观测站的气象观测变量之间 CE 和解耦度量之间的关系¹⁹

¹⁹Laura Mack, Marvin Kähnert, and Norbert Pirk. “Probabilistic modelling of atmosphere-surface coupling with a copula Bayesian network”. In: *Environmental Modelling & Software* (2026). arXiv: 2509.11975 [physics.ao-ph]. URL: <https://arxiv.org/abs/2509.11975>.

CASE 4: Land-atmosphere coupling by University of Oslo

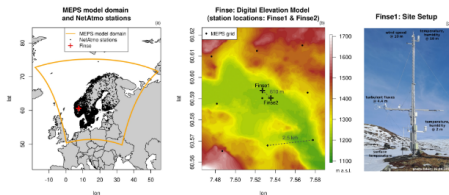


Figure 1: Used data sources: (a) Model domain of the NWP model MEPS and locations of the NetAtmo stations. (b) Surrounding of Finse and station locations. (c) Setup of the main site Finse1.

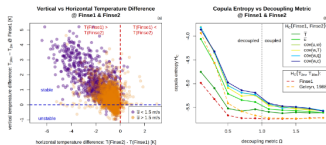


Figure 2: Relation of vertical and horizontal coupling: (a) Horizontal temperature difference between Finse1 and Finse2 versus vertical temperature difference at Finse1 colored based on a wind speed threshold (30 minutes temporal average), (b) decoupling metric at Finse1 versus copula entropy between Finse1 and Finse2 of temperature, wind speed, and the covariances corresponding to the fluxes of momentum $\text{cov}(u,w)$, sensible heat $\text{cov}(w,T)$, latent heat $\text{cov}(q,w)$ and CO_2 $\text{cov}(c,w)$ (30 minutes temporal average). Entropy calculation is based on time series of the same length to avoid a sampling size bias.

水文学 1

- 洪水预报
 - 金沙江流域洪水预报²⁰
 - 天山北麓奎屯河洪水分析²¹
- 河流相关性
 - 长江上游河段（金沙江、岷江、沱江、嘉陵江）相关性²²
 - 长江上游河段复合洪水事件分析²³
- 水沙关系分析
 - 黄河西柳沟河流域径流量和输沙量数据分析²⁴

²⁰ Lu Chen, Vijay P. Singh, and Shenglian Guo. "Measure of Correlation between River Flows Using the Copula-Entropy Method". In: *Journal of Hydrologic Engineering* 18.12 (2013), pp. 1591-1606, Lu Chen et al. "Copula entropy coupled with artificial neural network for rainfall-runoff simulation". In: *Stochastic Environmental Research and Risk Assessment* 28.7 (2014), pp. 1755-1767, 陈佳雷 et al. "一种基于多要素注意力时空图卷积网络的径流预报方法". Pat. CN111715128A, CN111715128A, 2023.

²¹Yingying Han et al. "Non-Stationary Flood Characteristics and Joint Risk Analysis in Inland China with Uncertainty Considerations". In: *Atmosphere* 17.3 (2026). p. 281.

²²Lu Chen and Shenglian Guo. *Copulas and its Application in Hydrology and Water Resources*. Springer Singapore, 2018.

²³Xu Wang and Yong-Ming Shen. "A Framework of Dependence Modeling and Evaluation System for Compound Flood Events". In: *Water Resources Research* 59.8 (2023). e2023WR034718. DOI: <https://doi.org/10.1029/2023WR034718>.

²⁴Longxia Qian et al. "A New Estimation Method for Copula Parameters for Multivariate Hydrological Frequency Analysis With Small Sample Sizes". In: *Water Resources Management* 36.4 (2022), pp. 1141–1157. ISSN: 1573-1650. URL: <https://doi.org/10.1007/s11269-021-03016-w>.

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水文学 4

- 水文观测网络选址和优化
 - 上海雨量观测网³⁸
 - 伊洛河流域水文观测网³⁹
 - 北京市区水文观测网; 汾河流域观测网; 太湖盆地流域雨量观测网⁴⁰
 - 淮河流域雨量观测网⁴¹
 - 美国查克托哈奇 (Choctawhatchee) 河流域水文观测网⁴²
- 气候变化和人类活动的水文系统响应⁴³

³⁸ Pengcheng Xu et al. "A two-phase copula entropy-based multiobjective optimization approach to hydrometeorological gauge network design". In: *Journal of Hydrology* 555 (2017), pp. 228–241.

³⁹ 王栋 et al. "一种基于 Copula 熵的水文站网优化模型的优化方法". Pat. CN106897530B. CN106897530B. 2019.

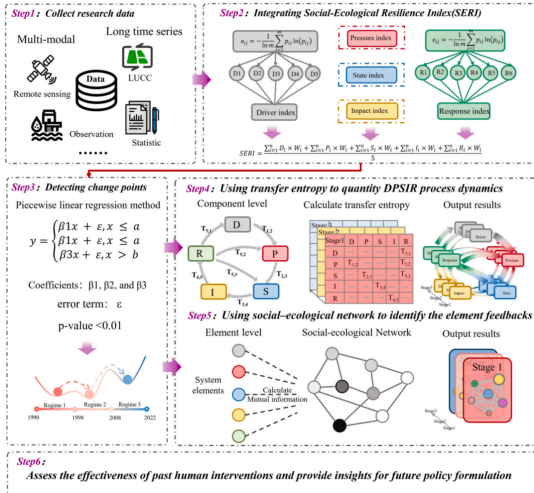
⁴⁰ Heshu Li et al. "Developing a dual entropy-transinformation criterion for hydrometric network optimization based on information theory and copulas". In: *Environmental Research* 180 (2020), p. 108813.

⁴¹ 徐鹏程 et al. "基于 C-Vine Copula 熵多目标优化模型的水文气象站网优化研究". In: *中国农村水利水电* 2 (2022), pp. 16–21. DOI: 10.12396/znsd.220702.

⁴² 杨惜岁. "多目标准则下流域水文站网的优化与评价". 硕士学位论文. 武汉理工大学, 2019.

⁴³ 蒋佩东. "基于 Bayesian 模型的长江流域年径流对变化环境响应的空间效应". 硕士学位论文. 武汉大学, 2023.

CASE 7: Social-ecological resilience in Karst region by CAS



Research methodology with copula entropy based transfer entropy

神经科学 1

● 神经病学

- 注意缺陷多动障碍⁶⁰
- 癫痫症⁶¹
- 脑卒中⁶²

⁶⁰Paolo Victor Redondo, Raphaël Huser, and Hernando Ombao. "Measuring information transfer between nodes in a brain network through spectral transfer entropy". In: *The Annals of Applied Statistics* 19.3 (2025), pp. 2386–2411. DOI: [10.1214/25-AOAS2050](https://doi.org/10.1214/25-AOAS2050). URL: <https://doi.org/10.1214/25-AOAS2050>, 汪方毅 et al. "基于静息态 fMRI 区分健康老年人认知水平的 MVPA 方法研究". In: *磁共振成像* 14.6 (2023), pp. 18–25.

⁶¹Zhaohui Li et al. "Estimating global phase synchronization by quantifying multivariate mutual information and detecting network structure". In: *Neural Networks* 183 (2025), p. 106984. ISSN: 0893-6080. DOI: <https://doi.org/10.1016/j.neunet.2024.106984>. URL: <https://www.sciencedirect.com/science/article/pii/S0893608024009134>.

⁶²Wojciech Ciezobka et al. "End-to-End Stroke Imaging Analysis Using Effective Connectivity and Interpretable Artificial Intelligence". In: *IEEE Access* 13 (2025), pp. 10227–10239. DOI: [10.1109/ACCESS.2025.3529179](https://doi.org/10.1109/ACCESS.2025.3529179).

神经科学 2

● 认知神经学

- 分析大脑认知活动的多模态数据⁶³
- 语音信息的编码和解析⁶⁴
- 神经元特异化识别⁶⁵

⁶³Stephanie J. Kayser et al. "Irregular Speech Rate Dissociates Auditory Cortical Entrainment, Evoked Responses, and Frontal Alpha". In: *The Journal of Neuroscience* 35.44 (2015), pp. 14691–14701, Robin A. A. Ince et al. "The Deceptively Simple N170 Reflects Network Information Processing Mechanisms Involving Visual Feature Coding and Transfer Across Hemispheres". In: *Cerebral Cortex* 26.11 (2016), pp. 4123–4135, Robin A.A. Ince et al. "A statistical framework for neuroimaging data analysis based on mutual information estimated via a gaussian copula". In: *Human Brain Mapping* 38.3 (2017), pp. 1541–1573, Etienne Combrisson et al. "Group-level inference of information-based measures for the analyses of cognitive brain networks from neurophysiological data". In: *NeuroImage* 258 (2022), p. 119347. ISSN: 1053-8119. DOI: <https://doi.org/10.1016/j.neuroimage.2022.119347>. URL: <https://www.sciencedirect.com/science/article/pii/S1053811922004669>.

⁶⁴Pieter De Clercq et al. "Beyond linear neural envelope tracking: a mutual information approach". In: *Journal of Neural Engineering* 20.2 (2023), p. 026007. DOI: 10.1088/1741-2552/acbe1d. URL: <https://dx.doi.org/10.1088/1741-2552/acbe1d>.

⁶⁵NA Pospelov et al. "Searching for cognitive specializations of neurons using mutual information framework". In: *Genes & Cells* 18.4 (2023), pp. 878–881.

神经科学 3

- 运动神经学
 - 分析运动的肌肉组合协同策略⁶⁶
- 计算神经学
 - 神经元可塑性建模⁶⁷
 - 神经网络信息传输关系分析⁶⁸

⁶⁶ 吴亚婷 et al. “多尺度肌间耦合网络分析”. In: 生物医学工程学杂志 38.4 (2021), pp. 742–752, Yating Wu et al. “R-Vine Copula Mutual Information for Intermuscular Coupling Analysis”. In: *Proceedings of the 11th International Conference on Computer Engineering and Networks*. 2022, pp. 526–534, David O’ Reilly and Ioannis Delis. “A network information theoretic framework to characterise muscle synergies in space and time”. In: *Journal of Neural Engineering* 19.1 (2022), p. 016031. DOI: 10.1088/1741-2552/ac5150. URL: <https://doi.org/10.1088/1741-2552/ac5150>, Shaojun Zhu et al. “Intermuscular coupling network analysis of upper limbs based on R-vine copula transfer entropy”. In: *Mathematical Biosciences and Engineering* 19.9 (2022), pp. 9437–9456, 金国美 et al. “基于小波包-Copula 互信息的肌间耦合特性”. In: 传感技术学报 35.10 (2022), pp. 1348–1353.

⁶⁷ Johannes Leugering and Gordon Pipa. “A Unifying Framework of Synaptic and Intrinsic Plasticity in Neural Populations.”. In: *Neural Computation* 30.4 (2018), pp. 945–986.

⁶⁸ Ari Pakman et al. “Estimating the Unique Information of Continuous Variables in Recurrent Networks”. In: *Advances in Neural Information Processing Systems* 34 (2021), pp. 20295–20307.

心理学

- 生物心理学
 - 情绪刺激下心跳诱发脑电位的时间交互现象⁶⁹

⁶⁹ Liesa Ravijts. "Revealing temporal interactions around the heartbeat-evoked potential modulated by emotional perception". MA thesis. Ghent Univeristy, 2019.

生物学

- 系统生物学
 - 生物信号调控和传导⁷⁰
 - 生物现象动态网络结构和功能⁷¹
- 生物信息学
 - 分析基因数据，研究生命和疾病机理⁷²
 - 筛选与癌症有关的变异基因⁷³
 - 单细胞测序基因调控网络构建⁷⁴

⁷⁰Agata Charzyńska and Anna Gambin. "Improvement of the k-NN Entropy Estimator with Applications in Systems Biology". In: *Entropy* 18.1 (2015), p. 13.

⁷¹Farzaneh Farhangmehr et al. "An information-theoretic algorithm to data-driven genetic pathway interaction network reconstruction of dynamic systems". In: *2013 IEEE International Conference on Bioinformatics and Biomedicine*. 2013, pp. 214–217.

⁷²Mario Wieser et al. "Inverse Learning of Symmetries". In: *Advances in Neural Information Processing Systems*. Vol. 33. 2020, pp. 18004–18015.

⁷³Qiang Wu and Dongxi Li. "CRIA: An Interactive Gene Selection Algorithm for Cancers Prediction Based on Copy Number Variations". In: *Frontiers in Plant Science* 13 (2022), p. 839044. ISSN: 1664-462X. DOI: 10.3389/fpls.2022.839044. URL: <https://www.frontiersin.org/article/10.3389/fpls.2022.839044>.

⁷⁴竺政彤. "基于单细胞测序数据构建基因调控网络的方法研究". 硕士学位论文. 内蒙古农业大学, 2023.

医学 2

● 临床医学 2

- 白内障术后角膜水肿风险预测⁷⁸
- 主动脉瓣置换手术射血分数分析⁷⁹
- 脑肿瘤医学影像组学诊断模型构建⁸⁰
- 基于脉搏波的高血压和糖尿病健康状态分类⁸¹

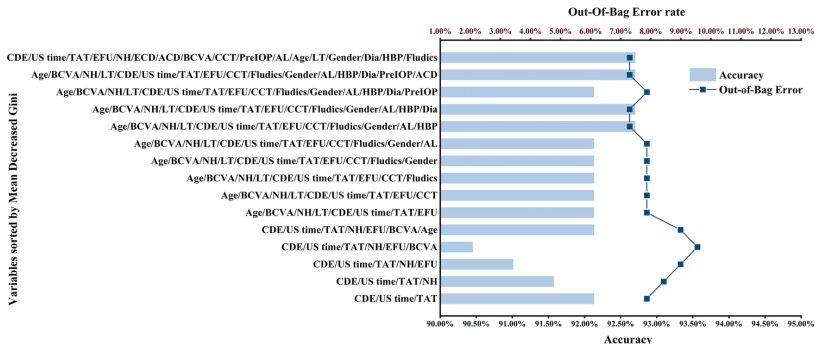
⁷⁸Yu Luo et al. "Research on Establishing Corneal Edema after Phacoemulsification Prediction Model Based on Variable Selection with Copula Entropy". In: *Journal of Clinical Medicine* 12.4 (2023), p. 1290. ISSN: 2077-0383. DOI: 10.3390/jcm12041290. URL: <https://www.mdpi.com/2077-0383/12/4/1290>.

⁷⁹S.M. Sunoj and N. Unnikrishnan Nair. "Survival Copula Entropy and Dependence in Bivariate Distributions". In: *REVSTAT-Statistical Journal* 23.1 (2025), pp. 101–115. URL: <https://revstat.ine.pt/index.php/REVSTAT/article/view/560>.

⁸⁰潘红宇. "基于影像组学与深度学习的脑肿瘤图像分类研究". 硕士学位论文. 西南大学, 2023.

⁸¹汤宇飞. "基于脉搏波的糖尿病和高血压诊断算法研究". 硕士学位论文. 中国矿业大学, 2023.

CASE 10: Corneal edema prediction by PLA 301 Hospital



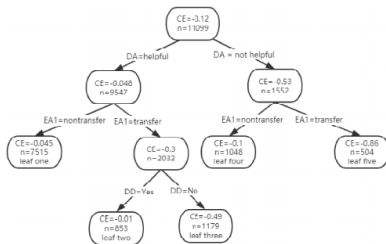
The predictive accuracy and out-of-bag error rate of each random forest model generated after selecting variables on the basis of copula entropy

药学

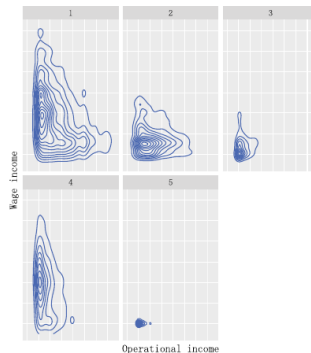
- 药物-靶点相互作用预测
 - 提出了一种交互式多特征融合的 DTIs 预测算法⁸⁷

⁸⁷ 高浩田 et al. “基于交互式多特征融合算法的药物靶标预测”. In: 太原理工大学学报 55.4 (2024), pp. 751–758.

CASE 11: Poverty policy evaluation by JXUFE



Binary decision tree by copula entropy



Joint density of two incomes of the families in each leaf

CASE 12: Commodity market analysis by UNESP



Figura 5.0.5: Transfer Entropy entre as séries da volatilidade de preços do açúcar e etanol.



Figura 5.0.6: Transfer Entropy entre as séries da volatilidade de preços do açúcar e gasolina.



Figura 5.0.7: Transfer Entropy entre as séries da volatilidade de preços de etanol e gasolina.

社会科学 3

- 教育学
 - 高中数学成绩与其他学科成绩相关性分析¹⁰¹
- 计算语言学
 - 城市热线派单系统知识图谱构建¹⁰²
 - 大模型词嵌入质量评估¹⁰³
- 新闻传播学
 - 上海新冠疫情下的公众情绪变化¹⁰⁴

¹⁰¹柳琼.“基于 Copula 和 MI 理论的相关性度量及其应用研究”. 硕士学位论文. 三峡大学, 2018.

¹⁰²陈作海, 钱恒, and 高永超. “一种基于知识图谱的城市热线派单方法及系统”. Pat. CN115860436A. CN115860436A. 2023.

¹⁰³Yanzhi Chen et al. "Neural Mutual Information Estimation with Vector Copulas". In: *Advances in Neural Information Processing Systems*. 2025.

¹⁰⁴ Bowen Zhang et al. "Changes in Public Sentiment under the Background of Major Emergencies – Taking the Shanghai Epidemic as an Example". In: *International Journal of Environmental Research and Public Health* 19.19 (2022), p. 12594. ISSN: 1660-4601. DOI: 10.3390/ijerph191912594. URL: <https://www.mdpi.com/1660-4601/19/19/12594>.

CASE 13: LLMs evaluation by University of Cambridge

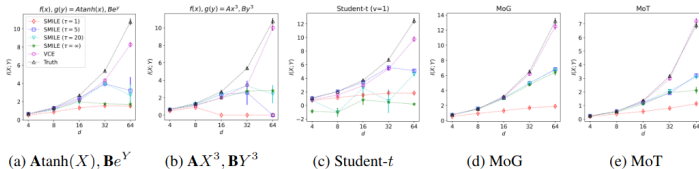


Figure 8: Comparison with the SMILE estimator under different clipping values τ .

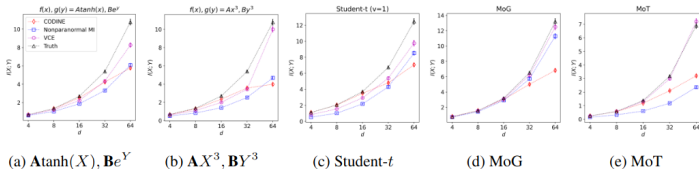


Figure 9: Comparing VCE with classic copula-based estimators e.g. nonparanormal MI (which uses a Gaussian copula to estimate MI) and CODINE (equivalent to classic copula transformation + MINE).

社会科学 4

- 法学
 - 社区属性与社区犯罪关系分析¹⁰⁵
- 政治学
 - 分析政权领导力因素和政权危机之间关系¹⁰⁶
- 军事学
 - 目标意图识别¹⁰⁷
- 情报学
 - 颠覆性技术科学-技术-产业互动模式分析¹⁰⁸
 - 高价值专利评估和发现¹⁰⁹

¹⁰⁵ Mario Wieser. "Learning Invariant Representations for Deep Latent Variable Models". PhD thesis. University of Basel, 2020.

¹⁰⁶ Stuart William Card. "Towards an Information Theoretic Framework for Evolutionary Learning". PhD thesis. Syracuse University, 2011.

¹⁰⁷ 张可 et al. "一种基于动态贝叶斯网络的目标意图识别方法". Pat. CN114997306A. CN114997306A. 2022.

¹⁰⁸ 许海云 et al. "颠覆性技术的科学-技术-产业互动模式识别与分析". In: 情报学报 42.7 (2023), pp. 816–831.

¹⁰⁹ 徐蕊婷, 陈哲宇, and 陈加鹏. "基于两阶段混合式特征选择的潜在高价值专利多情景判别方法研究". In: 数据分析与知识发现 (2026).

工学 1

● 能源工程 1

- 能源网络管理，研究天气因素与能源网络的耦合¹¹⁰
- 光伏发电功率预测¹¹¹
- 风电机组工况划分¹¹²
- 电力负荷预测¹¹³
- 风光储协同规划¹¹⁴

¹¹⁰ Xueqian Fu et al. "Uncertainty analysis of an integrated energy system based on information theory". In: *Energy* 122.122 (2017), pp. 649–662.

¹¹¹ 朱正林 和 张冕. "基于 AO 优化 VMD-CE-BiGRU 的光伏发电功率预测". In: *国外电子测量技术* 41.10 (2022), pp. 56–61.

¹¹² 崔双双 和 孙单勋. "分工况下风电机组各变量相关性研究". In: *综合智慧能源* 44.12 (2022), pp. 49–55.

¹¹³ Jian Ma. "Identifying Time Lag in Dynamical Systems with Copula Entropy based Transfer Entropy". In: *arXiv preprint arXiv:2301.06037*, arXiv:2301.06037 (2023). arXiv: 2301.06037 [cs.LG], Qin Yan et al. "Short-term prediction of integrated energy load aggregation using a bi-directional simple recurrent unit network with feature-temporal attention mechanism ensemble learning model". In: *Applied Energy* 355 (2024), p. 122159. ISSN: 0306-2619. DOI: <https://doi.org/10.1016/j.apenergy.2023.122159>. URL: <https://www.sciencedirect.com/science/article/pii/S0306261923015234>.

¹¹⁴ 董海艳 et al. "一种含源荷时序相似度约束的源储协同规划配置方法". Pat. CN114421538A. CN110766314A. 2022.

工学 2

● 能源工程 2

- 电网频率稳定性预测¹¹⁵
- 用户线损贡献分析¹¹⁶
- 配电网拓扑辨识¹¹⁷
- 电价预测¹¹⁸
- 电力系统宽频振荡影响因素和传播路径分析¹¹⁹
- 电力系统信息物理融合¹²⁰

¹¹⁵ Peili Liu et al. "Frequency Stability Prediction of Power Systems Using Vision Transformer and Copula Entropy". In: *Entropy* 24.8 (2022), p. 1165. ISSN: 1099-4300. DOI: 10.3390/e24081165. URL: <https://www.mdpi.com/1099-4300/24/8/1165>.

¹¹⁶ Wei Hu et al. "Research on User Loss Contribution Calculation of High-Loss Distribution Area Based on Transfer Entropy". In: *2022 China International Conference on Electricity Distribution (CICED)*. 2022, pp. 499–502. DOI: 10.1109/CICED56215.2022.9929052.

¹¹⁷ 秦超 and 潘毓笙. "一种基于时空特征的配电网拓扑辨识方法". Pat. CN117154679A. CN117154679A. 2023.

¹¹⁸ Xiaoping Xiong and Guohua Qing. "A hybrid day-ahead electricity price forecasting framework based on time series". In: *Energy* 264 (2022), p. 126099. ISSN: 0360-5442. DOI: <https://doi.org/10.1016/j.energy.2022.126099>. URL: <https://www.sciencedirect.com/science/article/pii/S0360544222029851>.

¹¹⁹ 冯双 et al. "一种电力系统宽频振荡影响因素和传播路径分析方法". Pat. CN114977222A. CN114977222A. 2022, 冯双 et al. "基于 Copula 传递熵的设备级和网络级宽频振荡传播路径分析及振荡源定位方法". In: *电工技术学报* 39.16 (2023), pp. 4996–5010. DOI: 10.19595/j.cnki.1000-6753.tces.230873.

¹²⁰ Fengtong Duan and Wanxing Sheng. "Conjugate Information-Energy Entropy Modeling and Entropy Gradient Control Analysis Based on Cyber-Physical Integration". In: *IEEE Transactions on Smart Grid* (2026), pp. 1–1.

CASE 14: Cyber-physical integration by State Grid

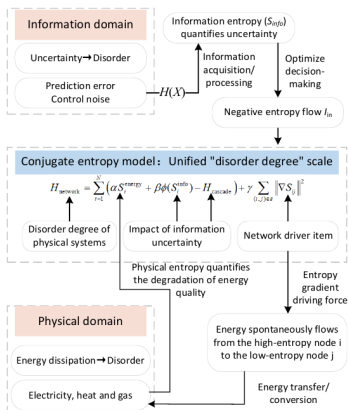


Fig. 1. The core concept diagram of the conjugate entropy theory.

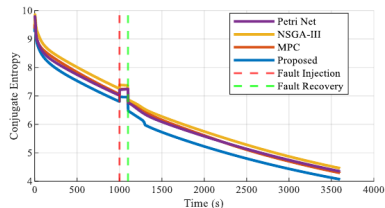


Fig. 7. Comparison of conjugate entropy.

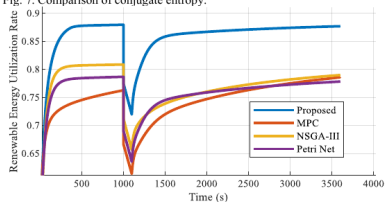


Fig. 8. Comparison of renewable energy utilization rate.

工学 4

- 纺织工程
 - 纱布生产过程质量控制¹²³
 - 可穿戴织物传感器方案设计¹²⁴
- 食品工程
 - 葡萄酒质量与理化成分关系分析¹²⁵

¹²³Seyedeh Azadeh Fallah Mortezaejad et al. "Non-Parametric Multivariate Control Chart Using Copula Entropy". In: *Sankhya B* 87 (2025), pp. 468–518. URL: <https://doi.org/10.1007/s13571-025-00374-y>.

¹²⁴Jiaqi Mo et al. "An Evaluation Framework for Textile Stretch Sensor Layouts Based on Mutual Information". In: *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing*. UbiComp Companion '25. Finland: Association for Computing Machinery, 2026, pp. 35–39. ISBN: 9798400714771. DOI: 10.1145/3714394.3754434. URL: <https://doi.org/10.1145/3714394.3754434>.

¹²⁵Marvin Lasserre, Régis Lebrun, and Pierre-Henri Willemin. "Learning Continuous High-Dimensional Models using Mutual Information and Copula Bayesian Networks". In: *Thirty-Fifth AAAI Conference on Artificial Intelligence, AAAI 2021*. AAAI Press, 2021, pp. 12139–12146. URL: <https://ojs.aaai.org/index.php/AAAI/article/view/17441>, Marvin Lasserre. "Apprentissages dans les réseaux bayésiens à base de copules non-paramétriques". PhD thesis. Sorbonne Université, 2022. URL: <https://tel.archives-ouvertes.fr/tel-03647090>.

CASE 15: Wine quality by Sorbonne Université

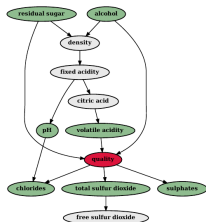


FIGURE 8.10 – Structure reconstruite avec l'algorithme B-CMIIC à partir de l'échantillon des vins rouges. Le paramètre α a été fixé à 0.04 en utilisant la méthode de *cross-validation*. La variable colorée en rouge est la variable cible tandis que l'ensemble des variables colorées en vert correspond à sa couverture de Markov.



FIGURE 8.11 – Structure reconstruite avec l'algorithme B-CMIIC pour l'échantillon des vins blancs. Le paramètre α est fixé à 0.02. La variable colorée en rouge est la variable cible tandis que l'ensemble des variables colorées en vert correspond à sa couverture de Markov.



FIGURE 8.12 – Structure reconstruite avec l'algorithme B-CMIIC pour l'échantillon des vins blancs. Le paramètre α est fixé à 0.06. La variable colorée en rouge est la variable cible tandis que l'ensemble des variables colorées en vert correspond à sa couverture de Markov.

工学 6

● 交通运输

- 大件货物运输方案制定¹³¹
- 航空和高铁票价影响因素分析¹³²
- 城市轨道交通客流分析和预测¹³³
- 铁路客流量预测¹³⁴

¹³¹ 黄达. “基于模块链构建的大件货物多式联运方案研究”. 博士学位论文. 北京交通大学, 2021.

¹³² 许罗豪 et al. “基于熵与回归树的票价影响因素研究”. In: *综合运输* 45.6 (2023), pp. 125–130.

¹³³ 王升. “基于多源数据的城市轨道交通系统客流分析与预测”. 硕士学位论文. 东南大学, 2022.

¹³⁴ Jiahao Chang and Xiaoyu Song. “A Railway Passenger Flow Prediction Model Based on Improved Prophet”. In: *Proceedings of the 2023 4th International Conference on Machine Learning and Computer Application*. 2024, pp. 798–804. doi: 10.1145/3650215.3650354, 常家豪. “基于集成模型的铁路客流量预测研究”. 硕士学位论文. 兰州交通大学, 2024.

工学 8

● 可靠性工程

- 系统退化过程建模¹⁴¹
- 风电机组健康状态评估¹⁴²
- 砂轮剩余寿命预测¹⁴³
- 滚动轴承剩余使用寿命预测¹⁴⁴

¹⁴¹Fuqiang Sun et al. "A Copula Entropy Approach to Dependence Measurement for Multiple Degradation Processes". In: *Entropy* 21.8 (2019), p. 724.

¹⁴²齐咏生 et al. "一种基于多维度 SCADA 数据评估风电机组健康状态评估方法". Pat. CN110442833A. CN110442833A. 2019.

¹⁴³程毅. "基于深度学习的砂轮剩余使用寿命预测". 硕士学位论文. 江南大学, 2023.

¹⁴⁴Zong Meng et al. "A remaining useful life prediction method of rolling bearings by RSA-BAFT combined with Copula Entropy feature selection". In: *Expert Systems with Applications* 275 (2025), p. 127100.

工学 9

- 石油工程
 - 煤层气井产量预测¹⁴⁵
 - 陆地原油生产碳排放分析¹⁴⁶
- 矿业工程
 - 煤-岩识别¹⁴⁷
 - 重介质选煤¹⁴⁸

¹⁴⁵ Zhifeng Luo and Haojiang Xi. "Hybrid Model Based on Copula Mutual Information and SSA-BP: Analysis of Key Factors and Prediction of Stable Gas Production". In: *Arabian Journal for Science and Engineering* 50 (2025), pp. 4673–4685. ISSN: 2191-4281. URL: <https://doi.org/10.1007/s13369-024-09205-0>.

¹⁴⁶ Zishang Yuan et al. "An analysis method of influencing factors of crude oil carbon footprint based on XGB-CE: a case study of an onshore oil production area in Shengli Oilfield, China". In: *Clean Technologies and Environmental Policy* 27.11 (2025), pp. 6573–6585. URL: <https://doi.org/10.1007/s10098-025-03203-y>.

¹⁴⁷ 高峰. "基于多源信息感知的煤-岩识别技术研究". 博士学位论文. 武汉大学, 2023.

¹⁴⁸ 建中华 and 代伟. "基于 Copula 熵优化的煤炭重介过程多变量时延参数估计方法". In: 2024 中国自动化大会. 2024.

工学 10

- 医学工程
 - 脑机接口¹⁴⁹
- 冶金工程
 - 真空蒸馏法制备高纯金属工艺参数优化¹⁵⁰

¹⁴⁹Chao Tang, Dongyao Jiang, and Badong Chen. "MEG Channel Selection Using Copula Entropy-Based Transfer Entropy for Motor Imagery BCI". In: *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. 2024, pp. 1–4. DOI: 10.1109/EMBC53108.2024.10782066, Chao Tang et al. "Copula Transfer Entropy-Based Channel Selection for MEG Motor Imagery Brain Computer Interfaces". In: *Tsinghua Science and Technology* 31.3 (2026), pp. 1474–1486. DOI: 10.26599/TST.2025.9010006. URL: <https://www.sciopen.com/article/10.26599/TST.2025.9010006>.

¹⁵⁰田庆华 et al. "一种真空蒸馏制备高纯金属的优化方法及优化系统". Pat. CN117577229A. CN117577229A. 2024.

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工学 12

- 核工程
 - 核反应堆传感器故障与事故预防¹⁵⁵

¹⁵⁵ 邓志光 et al. "铅铋快堆多工况多传感器状态监测方法、装置和验证系统". Pat. CN121350688A. CN121350688A. 2026, Jiarong Gao et al. "Research on nuclear power plant sensor data reconstruction based on Seq2Seq-Mamba". In: *Annals of Nuclear Energy* 231 (2026), p. 112219. ISSN: 0306-4549. DOI: <https://doi.org/10.1016/j.anucene.2026.112219>. URL: <https://www.sciencedirect.com/science/article/pii/S0306454926001076>.

● 航空航天

- 飞行器总体参数分析和优化¹⁵⁶
- 飞行器操纵品质评估¹⁵⁷
- 卫星在轨健康状态监测¹⁵⁸
- 涡扇发动机健康状态监测¹⁵⁹
- 机场间航班延误因果关系分析¹⁶⁰
- 航班到达时间预测¹⁶¹

156 [Baby Alpettiyil Krishnankutty, Rajesh Ganapathy, and Paduthol Godan Sankaran](#). "Non-parametric estimation of copula based mutual information". In: *Communications in Statistics - Theory and Methods* 49.6 (2020), pp. 1513–1527. DOI: [10.1080/03610926.2018.1563180](https://doi.org/10.1080/03610926.2018.1563180). eprint: <https://doi.org/10.1080/03610926.2018.1563180>. URL: <https://doi.org/10.1080/03610926.2018.1563180>.

¹⁵⁷张曙光 and 李雨函. “一种基于 transformer 模型的飞行器操纵品质评估方法”. Pat. CN121998517A. CN121998517A. 2026.

158 Hao Liu et al. "Data-driven identification model for associated fault propagation path". In: *Measurement* 188 (2022), p. 110628. ISSN: 0263-2241. DOI: <https://doi.org/10.1016/j.measurement.2021.110628>. URL: <https://www.sciencedirect.com/science/article/pii/S02632242121014986>, Zefan Zeng et al. "Satellite Telemetry Data Anomaly Detection Using Causal Network and Feature-Attention-Based LSTM". In: *IEEE Transactions on Instrumentation and Measurement* 71 (2022), pp. 1–21. DOI: [10.1109/TIM.2022.3151930](https://doi.org/10.1109/TIM.2022.3151930).

¹⁵⁹ 贾如侠, “涡扇发动机故障预测及剩余寿命分析方法研究”, 硕士学位论文, 哈尔滨师范大学, 2023.

¹⁶⁰ 吴格 et al. “一种因果关系分析方法及装置”. Pat. CN110766314A, CN110766314A, 2020.

¹⁶¹Xiaowei Tang et al. "Two Stages of Arrival Aircraft: Influencing Factors and Prediction of Integrated Arrival Time". In: *Aerospace* 12.3 (2025), p. 250. DOI: 10.3390/aerospace12030250.

工学 14

- 兵器工程
 - 武器效能评估指标体系约简¹⁶²
- 车辆工程
 - CAN 总线入侵检测¹⁶³
 - 高速动车组关键部件轴承健康监测¹⁶⁴
 - 自动驾驶感知模块优化¹⁶⁵
- 控制工程
 - 航站楼室内温度控制¹⁶⁶

¹⁶² 陈爱真 et al. “一种效能评估指标体系的约简方法和系统”. Pat. CN117634946A. CN117634946A. 2024.

¹⁶³ Sheng Gao et al. “Attack Detection for Intelligent Vehicles via CAN- Bus: A Lightweight Image Network Approach”. In: *IEEE Transactions on Vehicular Technology* 72.12 (2023), pp. 16624–16636. DOI: 10.1109/TVT.2023.3296705.

¹⁶⁴ Jie Xu et al. “An Adaptive Multi-Signal Framework for Real-Time Fault Diagnosis of Rolling Bearings”. In: *IEEE Transactions on Instrumentation and Measurement* 74 (2025), pp. 1–16. DOI: 10.1109/TIM.2025.3541779.

¹⁶⁵ Yue Zhou, Jihui Ma, and Honghui Dong. “Weighted Copula Entropy for Structural Pruning in Long-Tailed Autonomous Driving Object Detection”. In: *Entropy* 28.3 (2026), p. 336.

¹⁶⁶ Zhiwei Li et al. “Passenger-centric model predictive control for indoor temperature to facilitate energy-efficient airport terminal”. In: *Building and Environment* 283 (2025), p. 113344. ISSN: 0360-1323. DOI: <https://doi.org/10.1016/j.buildenv.2025.113344>. URL: <https://www.sciencedirect.com/science/article/pii/S0360132325008200>.

工学 15

- 电子工程
 - 集成电路封装材料物理性能预测¹⁶⁷
- 通信工程
 - 通讯网络加密技术研究¹⁶⁸
 - 6G 网络语义通信技术研究¹⁶⁹
 - 游戏系统编解码技术¹⁷⁰
- 高性能计算
 - 高性能计算能源效率优化¹⁷¹
- 信息安全
 - 深度神经网络对抗攻击防御算法¹⁷²

¹⁶⁷ 刘勃,“基于机器学习的封装材料加速预测”,硕士学位论文,哈尔滨理工大学,2022.

168 Xu Wang et al. "Physical Layer Secret Key Capacity Using Correlated Wireless Channel Samples". In: *2016 IEEE Global Communications Conference (GLOBECOM)*. 2016. pp. 1–6.

¹⁶⁹ 傅宇舟 et al., “面向 6G 网络的基于语义通信的端到端服务框架”, In: 移动通信 47.6 (2023), pp. 35–40.

170 Alexandre Delattre, Sylvain Gaeremynck, and Mickaël Corroyer. "Compressing entropy tables with interpolative coding". *Pat. US12341538B1*. US12341538B1. 2024.

¹⁷¹ [Andreas Gocht-Zech](#). "Ein Framework zur Optimierung der Energieeffizienz von HPC-Anwendungen auf der Basis von Machine-Learning-Methoden". PhD thesis. Technische Universität Dresden. 2022.

¹⁷²Lin Liu, Cong Hu, and Xiao-Jun Wu. “CE²: A Copula Entropic Mutual Information Estimator for Enhancing Adversarial Robustness”. In: *Pattern Recognition and Computer Vision*. Springer Singapore, 2024, pp. 163–174.

工学 16

- 测绘遥感
 - 高光谱遥感数据分析¹⁷³
 - 大型工程变形体内部监测¹⁷⁴
- 海洋工程
 - 海洋底质信息探测¹⁷⁵

¹⁷³ Xuexing Zeng and T S Durrani. "Band selection for hyperspectral images using copulas-based mutual information". In: 2009 IEEE/SP 15th Workshop on Statistical Signal Processing. 2009, pp. 341–344. DOI: 10.1109/SSP.2009.5278570.

¹⁷⁴ 曹久慧 et al. "一种基于自注意力机制的变形监测方法". Pat. CN116378120A. CN116378120A. 2023, 张旭辉. "顾及邻近点相似性的机器学习变形分析方法研究". 硕士学位论文. 武汉大学, 2022.

¹⁷⁵ 赵廷. "多波束声呐海底底质智能分类关键技术研究". 博士学位论文. 哈尔滨工程大学, 2022.

金融工程 1

- 投资组合优化
 - 股票资产相关性网络分析¹⁷⁶
 - ST 股票分类¹⁷⁷
- 金融问题建模
 - Copula 函数模型选择¹⁷⁸
- 股票相关性建模
 - R-vine copula 结构建模¹⁷⁹

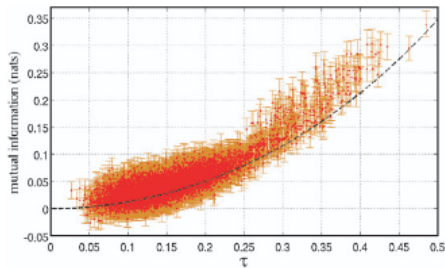
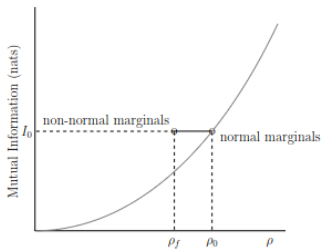
¹⁷⁶ Qitong Wang. "Social Networks, Asset Allocation and Portfolio Diversification". MA thesis. University of Waterloo, 2015, 廖轶楠. "基于 Copula 熵选股及集成神经网络预测的投资组合管理研究". 硕士学位论文. 南京信息工程大学, 2023.

¹⁷⁷ 朱仲儿. "多种机器学习方法的股票分类预测". 硕士学位论文. 上海师范大学, 2022.

¹⁷⁸ Rafael Calsaverini and Renato Vicente. "An information-theoretic approach to statistical dependence: Copula information". In: *EPL (Europhysics Letters)* 88.6 (2009), p. 68003. DOI: 10.1209/0295-5075/88/68003. URL: <https://doi.org/10.1209/0295-5075/88/68003>.

¹⁷⁹ Fadhah Amer Alanazi. "Truncating Regular Vine Copula Based on Mutual Information: An Efficient Parsimonious Model for High-Dimensional Data". In: *Mathematical Problems in Engineering* 2021 (2021), p. 4347957. URL: <https://doi.org/10.1155/2021/4347957>, 王念鸽. "基于互信息的 Vine Copula 模型的高频数据投资组合风险测度研究". 硕士学位论文. 浙江财经大学, 2023.

CASE 17: Copula modeling by Universidade de São Paulo





金融工程 3

- 金融产品价格预测
 - 基于因果关系的迁移学习价格预测模型构建¹⁸⁵
- 疫情对金融市场的影响
 - 疫情对股票指数相关性的影响¹⁸⁶
- 金融市场预测
 - 加密货币市场预测¹⁸⁷

¹⁸⁵Dabin Zhang et al. "A novel deep transfer learning framework with adversarial domain adaptation: application to financial time-series forecasting". In: *Neural Computing and Applications* 35 (2023), pp. 24037–24054. URL: <https://doi.org/10.1007/s00521-023-09047-1>.

¹⁸⁶Henryk Gurgul and Robert Syrek. "Mutual information between Polish subindexes –the use of copula entropy around the time of the COVID-19 pandemic". In: *Statistics in Transition new series* 25.1 (2024), pp. 23–41. DOI: 10.59170/stattrans-2024-002.

¹⁸⁷Amanda Mahmutovic. "Forecasting Cryptocurrency Returns In Adaptive Markets With an Extended Copula Based Feature Selection and Extended Copula Divergence Hybrid Loss Function". MA thesis. Linnaeus University, 2025.

Softwares

Third-party

- R: `cylcop`¹⁸⁹ (by University of Zürich and University of Minnesota);
- Python: `MLFinLab`¹⁹⁰ and `ArbitrageLab`¹⁹¹ (authored by Prof. Marcos López de Prado from Cornell University, Github 4500+ and 670+ stars);

¹⁸⁹Florian H. Hodel and John R. Fieberg. "cylcop: An R Package for Circular-Linear Copulae with Angular Symmetry". In: *bioRxiv* (2021), p. 2021.07.14.452253, Florian Hodel. *cylcop: Circular-Linear Copulas with Angular Symmetry for Movement Data*. CRAN. R package version 0.2.0, URL: <https://cran.r-project.org/package=cylcop>. 2022.

¹⁹⁰Hudson and Thames. *Machine Learning Financial Laboratory (MLFinLab)*. GitHub. 2021. URL: <https://github.com/hudson-and-thames/mlfinlab>.

¹⁹¹Hudson and Thames. *The ArbitrageLab package in Python*. GitHub. 2024. URL: <https://github.com/hudson-and-thames/mlfinlab>.

Third-party

- Python: **gcml**¹⁹² (by Glasgow University, Radboud University Nijmegen, University of Amsterdam, Kyiv AI group, Github 100+ stars), **pytorch-mighty**¹⁹³ (by Kyiv AI group), **HOI**¹⁹⁴ and **Frites**¹⁹⁵ (by CNRS, INT, and Aix-Marseille Université), **Tensorpac**¹⁹⁶ (by Université de Montréal, INT, and Aix-Marseille Université), **driada**¹⁹⁷ (by Moscow State University);

¹⁹²Robin A.A. Ince et al. "A statistical framework for neuroimaging data analysis based on mutual information estimated via a gaussian copula". In: *Human Brain Mapping* 38.3 (2017), pp. 1541–1573, Robin A.A. Ince et al. *gcmt: Gaussian-Copula Mutual Information*. GitHub. URL: <https://github.com/robinince/gcmt>. 2020.

¹⁹³Danylo Ulianych. *The pytorch-mighty package in Python*. GitHub. 2023. URL: <https://github.com/dizcza/pytorch-mighty>.

¹⁹⁴Brainets. *The HOI package in Python*. Github. 2024. URL: <https://brainets.github.io/hoi/>

¹⁹⁵Etienne Combrisson et al. "Frites: A Python package for functional connectivity analysis and group-level statistics of neurophysiological data". In: *Journal of Open Source Software* 7.79 (2022), p. 3842. DOI: 10.21105/joss.03842. URL: <https://doi.org/10.21105/joss.03842>.

¹⁹⁶[Etienne Combrisson et al. "Tensorpac: An open-source Python toolbox for tensor-based phase-amplitude coupling measurement in electrophysiological brain signals". eng. In: *PLoS computational biology* 16.10 \(2020\), e1008302. ISSN: 1553-7358. DOI: 10.1371/journal.pcbi.1008302.](#)

¹⁹⁷ [Institute for Advanced Brain Studies. *driada*: Dimensionality Reduction for Integrated Activity Data. Github. URL: https://github.com/iabs-neuro/driada. 2024.](https://github.com/iabs-neuro/driada)

- **Python: CopulaGP**¹⁹⁸ (by University of Edinburgh), **Polars-ds**¹⁹⁹ (by New York University, 1.45+ million downloads, Github 650+ stars), **copul**²⁰⁰ (by Albert-Ludwigs-Universität Freiburg), **rcopula**²⁰¹, and **pycopcor**²⁰² (by Technische Universität Dresden);

²⁰²Andreas Gocht-Zech. *The pycopcor package in Python*. Github. URL: <https://github.com/AndreasGocht/pycopcor>. 2023.

Third-party

- **Julia:** `CopEnt.jl`²⁰³ (by University of Edinburgh), `CausalityTools.jl`²⁰⁴ (by JuliaDynamics), and `Copulas.jl`²⁰⁵ (by Aix-Marseille Université, Github 110+ stars);
- **Matlab:** `gcmi`²⁰⁶ (by Glasgow University, Radboud University Nijmegen, University of Amsterdam, Kyiv AI group, Github 100+ stars), and `FieldTrip`²⁰⁷ (by Radboud University Nijmegen);
- **C++:** `NPStat`²⁰⁸ ²⁰⁹ (by Texas Tech University).

²⁰⁴Kristian Agasøster Haaga. *The CausalityTools.jl package v2.10.1 in Julia*. Github. 2023. URL: <https://github.com/JuliaDynamics/CausalityTools.jl>.

²⁰⁵Oskar Lavrny and Santiago Jimenez. "Copulas.jl: A fully Distributions.jl-compliant copula package". In: *Journal of Open Source Software* 9.94 (2024), p. 6189. DOI: 10.21105/joss.06189. URL: <https://doi.org/10.21105/joss.06189>.

²⁰⁶ Robin A.A. Ince et al. "A statistical framework for neuroimaging data analysis based on mutual information estimated via a gaussian copula". In: *Human Brain Mapping* 38.3 (2017), pp. 1541–1573, Robin A.A. Ince et al. *gcmi* : Gaussian-Copula Mutual Information. GitHub. URL: <https://github.com/robinince/gcml>. 2020.

²⁰⁷ Robert Oostenveld et al. "FieldTrip: Open Source Software for Advanced Analysis of MEG, EEG, and Invasive Electrophysiological Data". In: *Computational Intelligence and Neuroscience* 2011.1 (2011), p. 156869. DOI: <https://doi.org/10.1155/2011/156869>.

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²⁰⁹The CE estimation method is decomposed into two functions for empirical copula estimation and entropy estimation.

Monograph

Jian Ma. “Copula Entropy: Theory and Applications”. In: *arXiv preprint arXiv:2025.18168* (2025)

Thanks!

Q&A